Submodule 36.05: Multi-Layered Neural Networks (MLP) (Hinglish)

1. Single Perceptron ki Kamzori

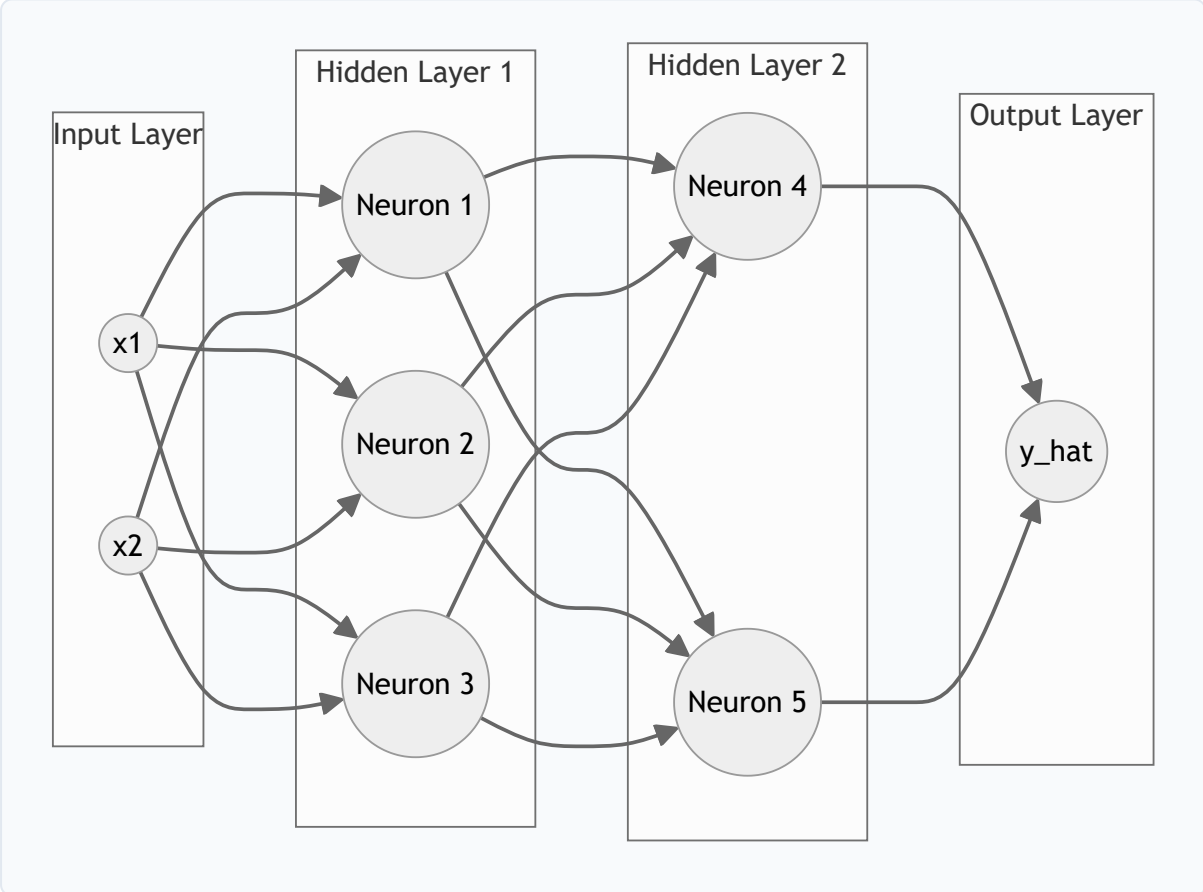
1969 mein researchers ne prove kiya ki ek akela Perceptron sirf wahi problems solve kar sakta hai jo ek seedhi line se separate ho sakein (**linearly separable** jaise AND, OR gates). Lekin jab baat **XOR gate** ya real-world complex problems ki aati hai jahan seedhi line nahi khichi ja sakti, wahan single perceptron fail ho jata hai.

Is problem ko solve karne ke liye **Multi-Layer Perceptron (MLP)** ka invention hua.

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2. Multi-Layer Neural Network ka Structure

Multi-layer network mein Input aur Output ke beech ek ya ek se zyada **Hidden Layers** laga di jati hain:



Zaroori Terms:

- **Fully Connected (Dense) Layer**: Aisi layer jisme pichli layer ka har ek neuron agli layer ke har ek neuron se juda hota hai.
- **Depth**: Network mein kitni layers hain. Jitni zyada layers, utna "Deep" network.
- **Width**: Ek layer ke andar kitne neurons hain.
- **Universal Approximation Theorem**: Yeh theorem kehta hai ki agar neural network mein non-linear activation ho, toh wo duniya ke kisi bhi complex function ya curve ko approximate (learn) kar sakta hai.

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3. Hidden Layers Complex Problems kaise solve karti hain?

Pehli layer data ki choti-choti lines banati hai, doosri layer un lines ko jodkar polygons aur curves banati hai, aur aakhri layer kisi bhi complex non-linear pattern (jaise XOR) ko aasani se solve kar leti hai.